

## 1 Question (15 points)

Given an array of integers that are greater than zero as a parameter, write a function called **splitIndex** that checks if it is possible to split this array in two (without changing the order of the elements), such that the sum of the two resulting arrays is the same. The method returns the splitting index (starting index of the second array) if it is possible to split the array; -1 otherwise.

```
public static int splitIndex(int[] A)
```

```
A = {5, 2, 3}
```

Returns 1.

```
A = {2, 3, 2, 1, 1, 1, 2, 1, 1}
```

Returns 3.

```
A = {1, 1, 3}
```

Returns -1.

```
A = {1, 1, 3, 1}
```

Returns -1.

## 2 Question (15 points)

Write a program that will generate a sequence of numbers according to the following algorithm. Your program will take an integer  $n$ . If  $n$  is even, the program will divide it by 2. If  $n$  is odd, you will multiply it by 3 and add 1. You repeat this process until you get 1.

Enter number: 22

22 11 34 17 52 26 13 40 20 10 5 16 8 4 2 1

Enter number: 1024

1024 512 256 128 64 32 16 8 4 2 1

### 3 Question (20 points)

Write a method that takes a matrix as parameter (2 dimensional array) and returns true if the matrix is a row increasing matrix, false otherwise. A matrix is row increasing if the numbers in each row are in increasing order.

## 4 Question (15 points)

Write a recursive method

```
int S(int N, int k)
```

to calculate Stirling numbers of the first kind where

$$S(n, n) = 1$$

$$S(n, 0) = 0, \text{ if } n \neq 0$$

$$S(n, 0) = 1, \text{ if } n = 0$$

$$S(n, k) = kS(n-1, k) + S(n-1, k-1)$$

## 5 Question (20 points)

Write a program that prompts the user to enter the number of websites, the domain names of each websites (e.g. boredpanda.com, onedio.com, sosyalon.com), and the number of visitors of these websites (e.g. 2255, 3578, 1185). Then, the program prints the list into an output file that has the websites in *decreasing order of their number of visitors*.

```
Enter number of websites: 3
```

```
Enter the name of website: boredpanda.com
```

```
Enter the number of visitors: 2255
```

```
Enter the name of website: onedio.com
```

```
Enter the number of visitors: 3578
```

```
Enter the name of website: sosyalon.com
```

```
Enter the number of visitors: 1185
```

```
Top Websites:
```

```
1) onedio.com 3578
```

```
2) boredpanda.com 2255
```

```
3) sosyalon.com 1185
```

## 6 Question (15 points)

Write a function **insert**, which takes a sorted array  $A$  and a value  $k$  as arguments, and returns a new array which contains all values of array  $A$  and the value  $k$  in a sorted manner. You are not allowed to use any sort function in this problem.

$A = \{2, 4, 5, 6\}$ ,  $k = 3$ .

$\{2, 3, 4, 5, 6\}$

$A = \{3, 4, 8\}$ ,  $k = 10$ .

$\{3, 4, 8, 10\}$